THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE MCQ TEST SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The exponent of 3 in the prime factorisation of 2250 is: 1

(a) 2 **(b)** 5 **(c)** 3 **(d)** 4

2. The number of zeroes of a polynomial whose graph cuts the x-axis at 1 point(s) is: 1

(a) 1 **(b)** 0 **(c)** 3 **(d)** 2

3. The pair of equations $3x + 2y = 5$ and $9x + 6y = 15$ is: 1

(a) no solution (parallel lines)
(b) exactly two solutions
(c) infinitely many solutions (coincident lines)
(d) exactly one solution (intersecting lines)

4. For the equation $x^2 - 7x + 12 = 0$, the sum of the roots is: 1

(a) 12 **(b)** 7 **(c)** -5 **(d)** -7

5. The 10th term of $21, 18, 15, 12, \ldots$ is: 1

(a) -3 **(b)** -6 **(c)** -9 **(d)** -30

6. In $\triangle ABC$, $DE \parallel BC$, $AD = 1$ cm, $DB = 3$ cm, $AE = 2$ cm. Then EC (in cm) $=$ 1

(a) 7 **(b)** 2 **(c)** 6 **(d)** 3

7. The distance between $(2, 3)$ and $(5, 7)$ is: 1

(a) 7 **(b)** 5 **(c)** 1 **(d)** 25

8. If $\tan A = \frac{4}{3}$, then $\sin A =$ 1

(a) $\frac{4}{3}$ **(b)** $\frac{4}{5}$ **(c)** $\frac{5}{4}$ **(d)** $\frac{3}{5}$

9. From a point on level ground, 15 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) 15 m **(b)** $15\sqrt{3}$ m **(c)** $5\sqrt{3}$ m **(d)** $30\sqrt{3}$ m

10. If tangents PA and PB from an external point P to a circle with centre O are inclined to each other at 80°, then $\angle POA$ is: 1

(a) 50° **(b)** 100° **(c)** 40° **(d)** 80°

11. A circular field has radius 14 m. Fencing costs ₹ 10 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 440 **(b)** ₹ 6160 **(c)** ₹ 88 **(d)** ₹ 880

12. A toy is a cone of base radius 7 cm and height 24 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 836 cm² **(b)** 550 cm² **(c)** 1012 cm² **(d)** 858 cm²

13. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 3, 16, 20, 4, 7 respectively, The mean of the data is: 1

(a) 44.2 **(b)** 45.2 **(c)** 442 **(d)** 45

14. Two dice are thrown simultaneously. The probability that the sum of the numbers is 8 is: 1

(a) $\frac{1}{9}$ **(b)** $\frac{5}{36}$ **(c)** $\frac{31}{36}$ **(d)** $\frac{1}{6}$

15. If HCF(306, 657) = 9, then LCM(306, 657) is: 1

(a) 201042 **(b)** 2754 **(c)** 963 **(d)** 22338

16. If one zero of $x^2 - 7x + 10$ is 2, the other zero is: 1

(a) 5 **(b)** -5 **(c)** 9 **(d)** 10

17. The solution of $x + y = 8$, $x - y = 2$ has $x =$ 1

(a) 5 **(b)** 3 **(c)** -5 **(d)** 8

18. For $x^2 - 6x + k = 0$ to have equal roots, $k =$ 1

(a) 3 **(b)** 36 **(c)** 9 **(d)** 6

19. The 5th term from the end of the AP with last term 100 and $d = 4$ is: 1

(a) 84 **(b)** 80 **(c)** 88 **(d)** 116

20. $\triangle ABC \sim \triangle DEF$ with areas 16 cm² and 36 cm². If $AB = 6$ cm, then DE (cm) $=$ 1

(a) 9 **(b)** 27 **(c)** 12 **(d)** 6

21. The point which divides the join of $(0, 0)$ and $(6, 9)$ in the ratio 1:2 internally is: 1

(a) $(2,\ 3)$ **(b)** $(6,\ 9)$ **(c)** $(3,\ \frac{9}{2})$ **(d)** $(4,\ 6)$

22. If $\cos A = \frac{5}{13}$, then $\tan A =$ 1

(a) $\frac{12}{5}$ **(b)** $\frac{12}{13}$ **(c)** $\frac{5}{12}$ **(d)** $\frac{13}{5}$

23. A ladder 6 m long leans against a vertical wall and makes an angle of 60° with the level ground. The height reached on the wall is: 1

(a) $6\sqrt{3}$ m **(b)** $\frac{3\sqrt{3}}{2}$ m **(c)** 3 m **(d)** $3\sqrt{3}$ m

24. A point P is 34 cm from the centre O of a circle of radius 16 cm. The length of the tangent drawn from P to the circle is: 1

(a) 30 cm **(b)** 900 cm **(c)** 18 cm **(d)** 50 cm

25. A wheel has radius 21 cm. The number of complete revolutions it makes to cover 66 m is: 1

(a) 100 **(b)** 25 **(c)** 50 **(d)** 60

26. A capsule is a cylinder of radius 3.5 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 374 cm² **(b)** 297 cm² **(c)** 748 cm² **(d)** 451 cm²

27. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 with frequencies 4, 3, 2, 4, 8, 4 respectively, The median of the data is: 1

(a) 73.4 **(b)** 80 **(c)** 78.75 **(d)** 70

28. A coin is tossed three times. The probability of getting exactly two heads is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{1}{2}$

29. The product of a non-zero rational and an irrational number such as $2\sqrt{3}$ is: 1

(a) always rational **(b)** always an integer

(c) sometimes rational **(d)** always irrational

30. A quadratic polynomial whose zeroes are 3 and 4 is: 1

(a) $x^2 + 7x + 12$ **(b)** $x^2 - 7x + 12$ **(c)** $x^2 + 7x - 12$ **(d)** $x^2 - 7x - 12$

31. The solution of $2x + 3y = 12$, $5x + 4y = 23$ has $y =$ 1

(a) 5 **(b)** 3 **(c)** 2 **(d)** -2

32. If one root of $x^2 - 9x + k = 0$ is 2, then k is: 1

(a) 9 **(b)** 16 **(c)** 14 **(d)** 7

33. If $a_n = 3n + 2$, then the common difference is: 1

(a) 3 **(b)** 2 **(c)** 6 **(d)** 5

34. The length of the diagonal of a square of side 3 cm, squared, is: 1

(a) 9 **(b)** 18 **(c)** 36 **(d)** 6

35. The area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 6)$ is: 1

(a) 5 **(b)** 10 **(c)** 24 **(d)** 12

36. The value of $\sin^2 30^\circ + \cos^2 30^\circ$ is: 1

(a) 1 **(b)** $\frac{1}{4}$ **(c)** 2 **(d)** $\frac{3}{2}$

37. A boy whose eyes are 2 m above the ground stands 10 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) $12\sqrt{3}$ m **(b)** 10 m **(c)** 12 m **(d)** 13 m

38. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 9$ cm, $BC = 11$ cm and $CD = 7$ cm. The length of AD is: 1

(a) 27 cm **(b)** 5 cm **(c)** 9 cm **(d)** 13 cm

39. The radii of two circles are 19 cm and 9 cm. The radius of the circle whose circumference equals the sum of their circumferences is: 1

(a) 14 cm **(b)** 30 cm **(c)** 56 cm **(d)** 28 cm

40. A solid is a cube of edge 14 cm surmounted by a hemisphere of diameter 14 cm (the hemisphere rests on a face of the cube). The total surface area of the solid is: 1

(a) 1330 cm² **(b)** 1484 cm² **(c)** 1022 cm² **(d)** 1176 cm²

41. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90 with frequencies 9, 4, 10, 13, 4 respectively, The mode of the data is: 1

(a) 70 **(b)** 72.5 **(c)** 64.75 **(d)** 80

42. A letter is chosen at random from the word 'MATHEMATICS'. The probability that it is a vowel is: 1

(a) $\frac{5}{11}$ **(b)** $\frac{3}{11}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{7}{11}$

43. The decimal expansion of $\frac{13}{3125}$ terminates after how many places of decimals? 1

(a) 7 **(b)** 5 **(c)** 6 **(d)** 8

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44. If α, β are zeroes of $x^2 - 9x + 14$, then $\alpha + \beta$ is: 1

(a) 14 **(b)** -9 **(c)** 9 **(d)** 10

45. For the pair $x + y = 10$ and $x - y = 2$, the value of xy is: 1

(a) 26 **(b)** 48 **(c)** 24 **(d)** 10

46. The roots of $x^2 - 8x + 15 = 0$ are: 1

(a) 3 and 5 **(b)** -3 and -5 **(c)** 3 and -5 **(d)** -3 and 5

47. If $S_n = n^2 + 2n$, then $a_1 =$ 1

(a) 3 **(b)** 1 **(c)** 4 **(d)** 2

48. A man goes 15 m due east and then 20 m due north. His distance from the start (m) is: 1

(a) 35 **(b)** 625 **(c)** 25 **(d)** 5

49. If the midpoint of $(2, 3)$ and (x, y) is $(5, 6)$, then $x =$ 1

(a) 12 **(b)** 8 **(c)** $\frac{7}{2}$ **(d)** 3

50. $\tan 20^\circ \cdot \tan 70^\circ =$ 1

(a) 2 **(b)** 20 **(c)** 1 **(d)** 0

END OF QUESTION PAPER **Total: 50 Marks**

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